Retrieval-Augmented LLMs with Indonesian Clinical Trials Guidelines: A Comparative Study

Ella Raputri
Computer Science Department
School of Computing and Creative Arts
Bina Nusantara University
Jakarta, Indonesia 11480
ella.raputri@binus.ac.id

Ari Jaya Teguh
Computer Science Department
School of Computing and Creative Arts
Bina Nusantara University
Jakarta, Indonesia 11480
ari.teguh@binus.ac.id

Saffanah Nur Hidayah
General Practitioner
Zamzam Clinic
South Jakarta, Indonesia 12550
saffanah.nh@gmail.com

Andari Karina Anom
Communication Department
School of Computing and Creative Arts
Bina Nusantara University
Jakarta, Indonesia 11480
aanom@binus.edu

Feri Setiawan
Computer Science Department
School of Computing and Creative Arts
Bina Nusantara University
Jakarta, Indonesia 11480
feri.setiawan@binus.edu

Nunung Nurul Qomariyah
Computer Science Department
School of Computing and Creative Arts
Bina Nusantara University
Jakarta, Indonesia 11480
nunung.qomariyah@binus.edu

2025 8th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI) | 979-8-3315-5768-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISRITI68345.2025.11393253

***Abstract*—Artificial Intelligence (AI), especially Large Language Model (LLM) powered chatbots, has emerged as a significant tool in our daily lives. Many industries, including the healthcare industry, nowadays use AI chatbots for their operational tasks. One of such tasks is for diagnosing diseases based on the patient's symptoms. However, in practice, LLM tends to generate incorrect information, so they need to be provided with more specific medical-related knowledge. Therefore, this study aims to enhance LLM diagnostic correctness by integrating Indonesian Clinical Trials Guidelines using RAG (Retrieval-Augmented Generation) to give the LLMs more context on medical cases. After that, we compare the diagnostic ability of three different RAG-enhanced popular LLMs (GPT, Deepseek, and Qwen). As a contrast, we also compare Deepseek model with and without RAG to test whether RAG leads to significant improvement for diagnosis correctness. Our research utilizes context-related metrics with ground truth from a general medical practitioner for evaluation. The result reveals that RAG significantly improves model performance and model type does not affect diagnosis correctness. However, from the mean alone, GPT has the highest performance with all metrics greater than 0.65, followed by Deepseek with RAG, Qwen, and Deepseek without RAG, but these differences were found to be not statistically significant. These findings further suggest that integrating local clinical guidelines with RAG can enhance medical reasoning in LLM, regardless of the underlying models, indicating that, as long as a reliable medical context is given, model type has a more limited impact on performance.**

***Index Terms*—Large Language Model (LLM), diagnosing diseases, RAG (Retrieval-Augmented Generation), context-related metrics**

I. Introduction

In recent years, Artificial Intelligence (AI) has become a topic that is intensively discussed in both research and real-world applications. Among all variations of AI, the AI chatbot powered by Large Language Model (LLM) has been a revolutionary one as it laid a foundation in creating intelligent agents that can respond to user requests using human-like languages [1]. This technological revolution can not be separated from the release of ChatGPT in 2022 [2], which deeply influenced the general public's view and curiosity in machine intelligence. Therefore, many industries started to take an interest in AI and adopt it in their operation processes.

Adoption of AI technology in various industries has been proven to increase productivity and creativity at the organizational level [3]. Among all industries, the integration of AI into the healthcare system has been a serious topic as many people now rely on AI chatbots for gathering information, including the diagnosis of diseases. According to the study in [4], more than 20 percent of health worker use ChatGPT for their health-related work. Reliance on AI chatbot, such as ChatGPT has become more prevalent these days as they can accelerate the diagnostic process, making it easier for patients and health workers to conclude a simple disease case [5]. Therefore, health workers can focus on more complex tasks that need their attention.

However, as with other technologies, AI also has its limitations. To date, many AI systems have been developed to help the healthcare industry. However, adoption and evaluation are still limited [6]. Besides that, AI chatbots tend to "hallucinate" or generate nonsensical information, so practical deployment of them to diagnose symptoms has also been a problem [7]. Hence, to enhance the reliability of the chatbot's response in a medical context, we combine LLMs with RAG (Retrieval-Augmented Generation) using information from PPPK (*Panduan Praktik Profesional Kedokteran*) or Indonesian Clinical Trials Guidelines. After that, we conduct evaluation to determine the performance of each model in the diagnostic process.

In conclusion, our research aims to compare the diagnostic ability of popular LLMs, such as Deepseek, Qwen, and GPT.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:51:38 UTC from IEEE Xplore. Restrictions apply.

To enhance every models' performance, we provide relevant medical context to them by using RAG. Therefore, it is hoped that each model can provide a more accurate response, and we can benchmark them to decide which one is the best in providing medical diagnoses.

II. Related Works

A. Large Language Models in Healthcare

Large Language Models (LLMs) are emerging in our daily lives, especially in the medical field. The AI model can determine a diagnosis through learning from large datasets of medical text, thus confirming their potential in the medical field [8]. The emerging LLMs influence is substantiated by the publications of 175 journals [9].

Goh [10] integrated LLMs into diagnostic workflows in random clinical trials, resulting in a more accurate and faster time to diagnose a case. Another study did a slight different way, by comparing two AI (GPT-3.5 and GPT-4) to see which model can generate a more accurate and reliable diagnosis. The outcome displayed the potential of LLMs in diagnostic support for clinics [11].

Regardless of what perks LLMs given, it still faces with issues with unreliability and accuracy, especially in high-risk scenario, where misdiagnosis can lead to fatality. A survey consisting of 550 studies displays both the possibilities of LLMs and also their drawbacks [12]. According to Chiu [13], in order to overcome the issues of accuracy and reliability of LLM, human oversight is still required to ensure there are no misdiagnoses, especially in a high-risk case.

B. Retrieval-Augmented Generation in Medical Context

Retrieval-Augmented Generation (RAG) is a method that combines external data resources like clinical guidelines, scientific literature databases, electronic health records, etc. Gargari also stated that RAG has become an important weapon to combat the challenges of stand-alone LLM in healthcare [14]. By providing external data resources that the models have not been programmed with, it also aids in preventing mislabeled information caused by human errors [15].

A study by Miao [16] displayed the benefits, which are improving the accuracy of medical outputs, on combining LLM and RAG in nephrology for diagnosis. A similar study by Bora [17] also exhibited the improvement on the outcome by integrating RAG-based databases and LLMs with medical text, especially where computing resources are scarce.

This has led to the generalizable benchmarking study that, in the field of medical retrieval-augmented generation, provides evaluation suites dedicated to the field that account for over 7,000 sets of clinical queries [18]. Statistically significant gains have been indicated by using RAG in the context of summarizing tasks, which increases the accuracy up to 6% for clinical information extraction as measured by the study in [19].

C. Evaluation Metrics for Medical AI Systems

BLEURT, a learned metric based on BERT, was trained on millions of synthetic instances and followed up by fine-tuning on human scores, in order to estimate coherent judgment even in the regime of small and out-of-distribution data [20].

In that sense, BERTScore is superior to n-gram based metrics as the comparison is performed by the tokens of the lens of contextual embeddings. These are the main characteristics that make BERTScore particularly convenient to use in situations where what counts most is the precision of meaning, as in the case of clinical use [21].

In the meantime, Semantic Answer Similarity (SAS) introduces a semantic assessment to question-answering task context because it employs a friendlier or harsher definition of the contact between question and answer. SAS has the capability to gauge outcomes comparable to those of human beings. This specialty is useful to assist in rating the medical responses [22].

In more recent times, there is G-EVAL allowing deployment of massive language models (GLM; GPT-4 or the like). While, on one hand, G-EVAL was observed closer to them in terms of human judgment for open-ended tasks empirically. However, in contrast, evaluating by using LLM could be biased [23].

D. Medical AI Evaluation in Non-English Languages

Even though the use of LLM in medical regions focuses more on the English language, other linguistic contexts are beginning to be recognized. In Indonesia, it is impossible to practice without local language skills. Culturalization of medical AI framework needs to address specific problems in the Indonesian context, as well as the translation process to some names or terms in the medical area.

When the research issues concerning medical AI assessment are considered through the perspective of Indonesia, a gap in the existing literature of a pronounced type is observed. The few existing literature cannot cover the huge field that LLM offers. So, it is critical to design the model so that the AI can still operate like normal even if the model is offered a non-English medical text.

III. Methodology

A. Dataset Overview and Description

The current dataset consists of 45 medical cases that have been carefully selected, in order to reflect a wide range of diagnostic situations of a typical clinical practice. In every case description, there is the exact syndrome of a patient in Indonesian. The dataset also includes the respective ground truth medical diagnoses verified by a competent licensed medical practitioner familiar with Indonesian medical language and Indonesian Clinical Trials Guidelines. Besides ground truth, the dataset also consists of the answer from each model. There are two different columns, which are the answer and the full answer. The answer column contains the diseases that the model predicts or diagnoses based on the medical cases. Meanwhile, the full answer column is the full answer from the model, including the conventional greetings and other

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:51:38 UTC from IEEE Xplore. Restrictions apply.

unnecessary information. The columns of the dataset can be seen in Table I.

Our dataset is obtained through sampling of our hospital partner (RSUD Gunung Jati Cirebon) SOAP data. SOAP, stands for Subjective, Objective, Assessment, and Plan, is a standardized method for hospital to document patient medication progress. To ensure representativeness, we sampled the medical cases based on the ICD-10 keys (international medical coding for disease classification) exists in the dataset, so that we can cover all the diseases. However, limitations and biases may still potentially arise. Some diseases may be unrepresented due to their nonexistence in the records. Using data only from a healthcare partner may also introduce bias due to patient demographics and local institutional practices.

B. Dataset Preprocessing

First of all, we prepared a set of sample questions that represent various medical cases. Then, we gather the answers from different LLMs by prompting them. To increase response accuracy, we use the RAG approach using the previously stored Indonesian Clinical Trials Guidelines document inside the AstraDB vector database. Therefore, the models can now use the document as their reference source when giving diagnoses.

After the data gathering step, we obtained the full diagnostic answer from each model. From the full answer of each model, we extract the predicted diseases. Then, we invited a medical practitioner to establish the ground truth for each sample medical case.

The doctor provides the diagnoses for the medical cases based on her experience. The result that she provided is the main disease (the most potential to be the correct answer for the question) and the differential diagnosis (other possible similar diseases with similar symptoms in the question).

C. Model and Techniques

1) LLM Selection: Three well-known LLMs were chosen to be tested under two conditions, one being a standalone mode (no external knowledge is supplemented) and another being enhanced mode, provided by RAG. The selected models were QWEN 2.5 72B, GPT-4o mini, and Deepseek-V3.

2) RAG Implementation: The RAG system is developed to incorporate information contained in the Indonesian Clinical Trials Guidelines as the sources of diagnostic knowledge. The retrieval module, which uses semantic similarity searching, then retrieves the relevant medical information based on the description of the patient symptoms.

The prompt that was used in the RAG was framed in Indonesian in order to bring out consistency of evaluated models. The Indonesian Clinical Trials Guidelines document and the question for the sample medical case itself is in Indonesian. Therefore, to generate a consistent Indonesian answer, we utilized Indonesian prompt.

D. Evaluation Techniques

Before we evaluate the candidate sentences, we first normalized them by matching them to a standardized synonyms that was listed on the synonym map. Therefore, words with similar meanings, but different wordings, such as ”demam dengue” and ”demam berdarah” would be standardized to become ”demam berdarah dengue”. The synonym maps were obtained by mapping similar diseases based on the previous SOAP records.

In practice, ”demam dengue” and ”demam berdarah dengue” are not the same, although they are both caused by the dengue virus. The difference is in the laboratory results. Based on Indonesian Clinical Trials Guidelines, ”demam dengue” can be a separate diagnosis due to the lack of ”demam berdarah” criteria. Meanwhile, a patient is diagnosed to have ”demam berdarah” if they already satisfy the ”demam berdarah” criteria. However, for our current research, we focused on more generalized results that are easier to comprehend by the public, so we standardized them to be the same.

Next, four metrics of evaluation were complemented to make a complete analysis of performance:

- BERT (Bidirectional Encoder Representations from Transformers) Score: Calculate the similarity score between the candidate sentence and ground truth based on the sum of cosine similarities between the pre-trained BERT contextual embeddings of their tokens [21].
- BLEURT: a trained evaluation metric based on BERT that conveys human-level judgment of reference versus candidate sentence. BLEURT is not calibrated (normalized), so we plotted a distribution graph based on BLEURT and used it to compare the model's performance [20].
- SAS (Semantic Answer Similarity): A cross-encoder-based evaluation metric that estimates the semantic meaning of the answer or candidate sentence and compare it with the reference sentence [22].
- LLM-based Judge or G-Eval: prompt-based evaluator that utilizes the GPT LLM family as the judge to simulate human quality judgment [23]. In our evaluation process, we utilized the GPT-4-Turbo.

After that, inferential statistical tests are conducted to evaluate the significance of each model score difference. There are two main hypotheses that we tested:

1) **Effect of RAG**: Model with RAG performs better than the model without RAG. Let $\overline{X}$ denote the mean score.
   $H_0 : \overline{X}_{\text{(Deepseek RAG)}} \leq \overline{X}_{\text{(Deepseek nonRAG)}}$
   $H_1 : \overline{X}_{\text{(Deepseek RAG)}} > \overline{X}_{\text{(Deepseek nonRAG)}}$
2) **Effect of model type**: The model type has a significant effect on the score. Let j denote the model type that uses RAG and β_j denote the effect (coefficient) score.
   $H_0 : \beta_j = 0$ for all j
   $H_1 : \exists j$, such that $\beta_j \neq 0$

E. Research Flow Overview

The overview of our research flow can be seen in Figure 1. We first started with defining the research problem and hypotheses. Next, we continue with documents gathering and RAG incorporation to models. After that, we sampled the medical cases from the hospital SOAP data and gathered

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:51:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I: Dataset's Details

Columns	Description	Data Type
No	the serial number of the medical case	float
Question	the medical case statement in the form of question	object (string)
Dr Answer	the ground truth for that medical case	object (string)
Qwen Answer	diseases diagnosed by Qwen for that medical case	object (string)
GPT Answer	diseases diagnosed by GPT for that medical case	object (string)
Deepseek RAG Answer	diseases diagnosed by Deepseek for that medical case with RAG	object (string)
Deepseek Non RAG Answer	diseases diagnosed by Deepseek for that medical case without using RAG	object (string)
Qwen Full Answer	full answer from Qwen	object (string)
GPT Full Answer	full answer from GPT	object (string)
Deepseek RAG Full Answer	full answer from Deepseek using RAG	object (string)
Deepseek Non RAG Full Answer	full answer from Deepseek without using RAG	object (string)

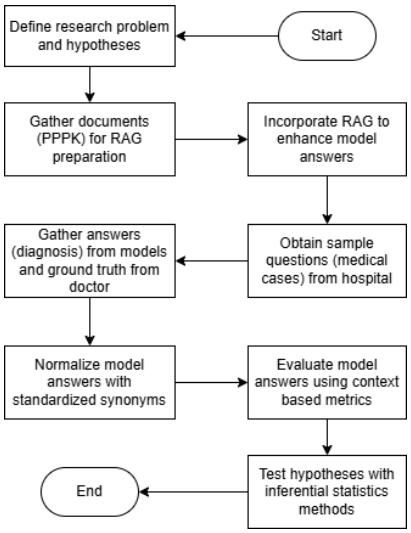


Fig. 1: Research Flow Flowchart

diagnoses from models and ground truth from the doctor. Lastly, we normalized each model's answer, then benchmarked the models. Hypothesis testing are conducted to evaluate the significancy of metrics difference between each model.

All three models in this research are presented with uniform RAG setups and the same sample questions. However, we also include an additional model, which is Deepseek without RAG as a baseline to compare RAG and non-RAG LLM diagnostic capability. Deepseek is chosen due to its cost-effectiveness over other models. The instructions and prompts involved in carrying out the assessments were also uniform, so that the evaluation is considered 'fair' for all models.

IV. Results and Discussion

Table II shows the descriptive statistics (mean and standard deviation) of the evaluation score for each model when they are benchmarked with the medical case dataset.

From Table II, we can see that by overall mean, GPT's performance is the best, followed by Deepseek with RAG, Qwen, and lastly, Deepseek without RAG. Deepseek without RAG rank as the last one, as it does not have any additional context given by RAG. From the table, it can also be observed that the standard deviation for both SAS and LLM-based judge scores is relatively high, suggesting that the values vary from very low to very high scores.

From the BERT scores, it can also be inferred that all LLMs have better recall than precision in diagnosing diseases. In other words, the models tend to include many relevant (positive) terms in their diagnosis, but are inclined to over-predicting positives. Hence, resulting in a lower precision score than the recall score. However, in a medical diagnosis case, a higher recall score is more favorable than a higher precision score, because it is better to have a false positive than a false negative, so that there is no disease that is missed in the diagnosis.

Next, we tested the hypotheses. A Shapiro-Wilk test is first conducted to test the data normality. The result can be seen in Table III. If $p < 5 \times 10^{-2}$, then the data is not normal. Based on the test, all SAS and LLM-based judge scores for each model are not normal, except LLM-based judge score for Deepseek without RAG. Therefore, for hypothesis testing that involves SAS or LLM-based judge scores, we will use non-parametric tests that do not require the data to be normal.

For the first main hypothesis, we tested each metric from Deepseek with RAG and without RAG with t-test for normalized data and Wilcoxon test for unnormalized data. The result can be seen in Table IV. Among 4 out of 5 metrics, the result showed a statistically significant improvement of using RAG, with SAS being the only exception.

For the second main hypothesis, we tested whether the model type has a significant effect on performance scores by comparing metrics from all models that are enhanced using RAG. The result can be seen in Table V. Unlike the first hypothesis, the result is more inconsistent, suggesting that model type may not have a significant effect for predicting the correct diagnosis for a disease.

Besides scores such as BERT, SAS, and LLM-based Judge, we also evaluate each model's performance with BLEURT. However, as stated in the Methodology part, the BLEURT score is not normalized, so we plotted the histogram to visualize the distribution of the BLEURT score of each model's diagnosis. The histogram can be seen in Figure 2.

The histograms in Figure 2 reveal that GPT has the best result because the distribution is skewed towards a positive score. However, it is left skewed, and still produces a high

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:51:38 UTC from IEEE Xplore. Restrictions apply.

TABLE II: LLM Performance Descriptive Statistics (Mean and Standard Deviation)

	GPT		Deepseek with RAG		Deepseek without RAG		Qwen	
Evaluation Metrics	**Mean**	σ	**Mean**	σ	**Mean**	σ	**Mean**	σ
BERT's Precision	0.820	0.043	0.812	0.039	0.791	0.05	0.814	0.035
BERT's Recall	0.874	0.053	0.875	0.042	0.843	0.044	0.850	0.058
BERT's F1	0.846	0.045	0.842	0.036	0.816	0.043	0.831	0.038
SAS	0.688	0.286	0.653	0.255	0.414	0.249	0.494	0.296
LLM-based Judge	0.755	0.204	0.758	0.227	0.691	0.289	0.616	0.286

TABLE III: Each Metrics Shapiro-Wilk Test Result

Evaluation Metrics	**GPT**	**Deepseek with RAG**	**Deepseek without RAG**	**Qwen**
BERT's Precision	6.18×10^{-1}	3.77×10^{-1}	2.09×10^{-1}	1.52×10^{-1}
BERT's Recall	7.56×10^{-2}	8.70×10^{-1}	5.35×10^{-1}	8.38×10^{-2}
BERT's F1	5.50×10^{-2}	3.89×10^{-1}	5.42×10^{-2}	5.15×10^{-1}
SAS	3.16×10^{-8}	4.17×10^{-9}	1.20×10^{-7}	8.13×10^{-7}
LLM-based Judge	1.51×10^{-4}	8.14×10^{-3}	2.23×10^{-1}	2.76×10^{-2}

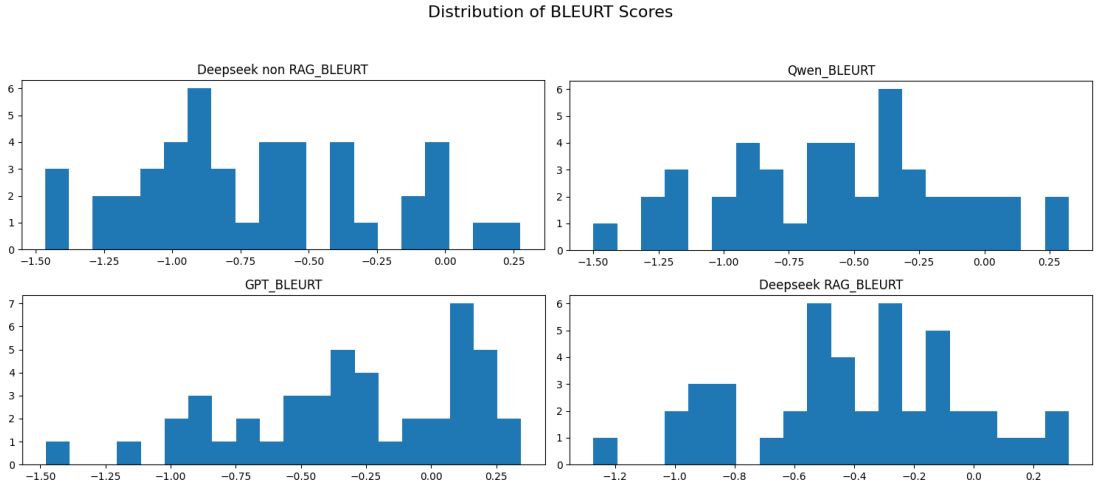


Fig. 2: BLEURT Distribution Histogram for Each LLM

TABLE IV: Hypothesis 1 Testing for Each Metrics

Evaluation Metrics	**Test Type**	**p-value**	**Conclusion**
BERT's Precision	t-test	4.37×10^{-3}	Reject H_0
BERT's Recall	t-test	3.81×10^{-5}	Reject H_0
BERT's F1	t-test	1.34×10^{-4}	Reject H_0
SAS	Wilcoxon	1.15×10^{-1}	Cannot reject H_0
LLM-based Judge	Wilcoxon	2.84×10^{-6}	Reject H_0

TABLE V: Hypothesis 2 Testing for Each Metrics

Evaluation Metrics	**Test Type**	**p-value**	**Conclusion**
BERT's Precision	ANOVA	2.74×10^{-1}	Cannot reject H_0
BERT's Recall	ANOVA	6.75×10^{-3}	Reject H_0
BERT's F1	ANOVA	5.56×10^{-2}	Cannot reject H_0
SAS	Friedman	3.05×10^{-4}	Reject H_0
LLM-based Judge	Friedman	1.41×10^{-2}	Reject H_0

negative score (about -1.50). Next are Deepseek with RAG and Qwen. Both histograms are similar, but Deepseek with RAG is slightly better than Qwen because it does not have a strongly negative value. Lastly, there is Deepseek without using RAG. The histogram shows that the model prediction is unstable, with some predictions being good (positive score), but some are strongly negative.

V. Conclusion and Future Work

In conclusion, this study analyzes the diagnosis correctness of each popular LLM by collaborating with a doctor for the ground truth establishment. Each model is also enhanced by RAG using the Indonesian Clinical Trials Guidelines to provide context related to disease diagnosis. Then, we evaluate each model using context-based metrics.

After reviewing the metrics and hypothesis tests, the result showed that there is a significant effect of using RAG to improve the model performance. However, our main finding is

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:51:38 UTC from IEEE Xplore. Restrictions apply.

that the model type does not have a significant effect for the correctness of medical diagnosis. Based on the overall mean score alone, the order of models is GPT, Deepseek with RAG, Qwen, and then followed by Deepseek without RAG, but the difference is not statistically significant. For clinicians, this finding provides an insight on practical deployment of the AI. It implies that providing relevant diagnosis knowledge is more important than model choices, thereby reliable diagnostic model can be achieved even with more cost-efficient models.

Looking ahead, future research could focus on evaluating more diverse models and datasets. We plan to gather more data to be used as the medical cases in the benchmark dataset. Besides that, collaborating with more medical practitioners to establish the ground truth for each medical case is also beneficial, so that we can obtain a more representative ground truth. Choosing a specific medical dataset for evaluation, such as a dataset that only contains heart diseases, can also be used to benchmark each model to determine which model is the best in diagnosing diseases in a specific medical field.

Supplementary Codes

All codes that were used in our research can be accessed in our GitHub repository through the link: https://github.com/Ella-Raputri/LLMDiagnosisPPPK.

Contributorship

Ella Raputri: Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing-Original Draft, Writing-Review & Editing, Visualization. **Ari Jaya Teguh**: Methodology, Software, Investigation, Writing-Original Draft. **Saffanah Nur Hidayah**: Validation, Resources. **Andari Karina Anom**: Resources. **Nunung Nurul Qomariyah, Feri Setiawan**: Conceptualization, Writing-Review & Editing, Supervision, Project Administration.

References

[1] S. Paliwal, V. Bharti, and A. K. Mishra, "Ai chatbots: Transforming the digital world," in *Recent trends and advances in artificial intelligence and internet of things*. Springer, 2019, pp. 455–482.

[2] F. Sun, "Chatgpt, the start of a new era," *A Bright and Gloomy Future. In https://feisun. org/2022/12/23/-chatgpt-the-start-of-a-new-era (ultima consultazione: 27/03/2023)*, 2022.

[3] H. Taherdoost and M. Madanchian, "Artificial intelligence and knowledge management: Impacts, benefits, and implementation," *Computers*, vol. 12, no. 4, p. 72, 2023.

[4] M.-H. Temsah, F. Aljamaan, K. H. Malki, K. Alhasan, I. Altamimi, R. Aljarbou, F. Bazuhair, A. Alsubaihin, N. Abdulmajeed, F. S. Alshahrani, R. Temsah, T. Alshahrani, L. Al-Eyadhy, S. M. Alkhateeb, B. Saddik, R. Halwani, A. Jamal, J. A. Al-Tawfiq, and A. Al-Eyadhy, "Chatgpt and the future of digital health: A study on healthcare workers' perceptions and expectations," *Healthcare*, vol. 11, no. 13, 2023. [Online]. Available: https://www.mdpi.com/2227-9032/11/13/1812

[5] I. Altamimi, A. Altamimi, A. S. Alhumimidi, A. Altamimi, and M.-H. Temsah, "Artificial intelligence (ai) chatbots in medicine: A supplement, not a substitute," *Cureus*, Jun. 2023. [Online]. Available: http://dx.doi.org/10.7759/cureus.40922

[6] S. Reddy, W. Rogers, V.-P. Makinen, E. Coiera, P. Brown, M. Wenzel, E. Weicken, S. Ansari, P. Mathur, A. Casey, and B. Kelly, "Evaluation framework to guide implementation of ai systems into healthcare settings," *BMJ Health amp; Care Informatics*, vol. 28, no. 1, p. e100444, Oct. 2021. [Online]. Available: http://dx.doi.org/10.1136/bmjhci-2021-100444

[7] Z. Ji, T. Yu, Y. Xu, N. Lee, E. Ishii, and P. Fung, "Towards mitigating LLM hallucination via self reflection," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 1827–1843. [Online]. Available: https://aclanthology.org/2023.findings-emnlp.123/

[8] S. Zhou, Z. Xu, M. Zhang, C. Xu, Y. Guo, Z. Zhan, Y. Fang, S. Ding, J. Wang, K. Xu, L. Xia, J. Yeung, D. Zha, D. Cai, G. B. Melton, M. Lin, and R. Zhang, "Large language models for disease diagnosis: a scoping review," *npj Artificial Intelligence*, vol. 1, 6 2025.

[9] D. Wang and S. Zhang, "Large language models in medical and healthcare fields: applications, advances, and challenges," *Artificial Intelligence Review*, vol. 57, 11 2024.

[10] E. Goh, R. Gallo, J. Hom, E. Strong, Y. Weng, H. Kerman, J. A. Cool, Z. Kanjee, A. S. Parsons, N. Ahuja, E. Horvitz, D. Yang, A. Milstein, A. P. Olson, A. Rodman, and J. H. Chen, "Large language model influence on diagnostic reasoning: A randomized clinical trial," *JAMA Network Open*, vol. 7, 10 2024.

[11] A. Ríos-Hoyo, N. L. Shan, A. Li, A. T. Pearson, L. Pusztai, and F. M. Howard, "Evaluation of large language models as a diagnostic aid for complex medical cases," *Frontiers in Medicine*, vol. 11, 2024.

[12] X. Meng, X. Yan, K. Zhang, D. Liu, X. Cui, Y. Yang, M. Zhang, C. Cao, J. Wang, X. Wang, J. Gao, Y. G. S. Wang, J. ming Ji, Z. Qiu, M. Li, C. Qian, T. Guo, S. Ma, Z. Wang, Z. Guo, Y. Lei, C. Shao, W. Wang, H. Fan, and Y. D. Tang, "The application of large language models in medicine: A scoping review," *iScience*, vol. 27, 5 2024.

[13] W. H. K. Chiu, W. S. K. Ko, W. C. S. Cho, S. Y. J. Hui, W. C. L. Chan, and M. D. Kuo, "Evaluating the diagnostic performance of large language models on complex multimodal medical cases," *Journal of Medical Internet Research*, vol. 26, 2024.

[14] O. K. Gargari and G. Habibi, "Enhancing medical ai with retrieval-augmented generation: A mini narrative review," 1 2025.

[15] R. Yang, Y. Ning, E. Keppo, M. Liu, C. Hong, D. S. Bitterman, J. C. L. Ong, D. S. W. Ting, and N. Liu, "Retrieval-augmented generation for generative artificial intelligence in health care," *npj Health Systems*, vol. 2, 1 2025.

[16] J. Miao, C. Thongprayoon, S. Suppadungsuk, O. A. G. Valencia, and W. Cheungpasitporn, "Integrating retrieval-augmented generation with large language models in nephrology: Advancing practical applications," 3 2024.

[17] A. Bora and H. Cuayáhuitl, "Systematic analysis of retrieval-augmented generation-based llms for medical chatbot applications," *Machine Learning and Knowledge Extraction*, vol. 6, pp. 2355–2374, 12 2024.

[18] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking retrieval-augmented generation for medicine," *Association for Computational Linguistics*, pp. 6233–6251, 8 2024. [Online]. Available: https://github.com/Teddy-XiongGZ/MedRAG https://aclanthology.org/2024.findings-acl.372/

[19] M. Alkhalaf, P. Yu, M. Yin, and C. Deng, "Applying generative ai with retrieval augmented generation to summarize and extract key clinical information from electronic health records," *Journal of Biomedical Informatics*, vol. 156, 8 2024.

[20] T. Sellam, D. Das, and A. P. Parikh, "Bleurt: Learning robust metrics for text generation," *Association for Computational Linguistics*, pp. 7881–7892, 6 2020. [Online]. Available: http://github.com/google-research/

[21] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "Bertscore: Evaluating text generation with bert," *International Conference on Learning Representations*, 2 2020. [Online]. Available: http://arxiv.org/abs/1904.09675

[22] J. Risch, T. Möller, J. Gutsch, and M. Pietsch, "Semantic answer similarity for evaluating question answering models," in *Proceedings of the 3rd Workshop on Machine Reading for Question Answering*, A. Fisch, A. Talmor, D. Chen, E. Choi, M. Seo, P. Lewis, R. Jia, and S. Min, Eds. Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 149–157. [Online]. Available: https://aclanthology.org/2021.mrqa-1.15/

[23] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, "G-eval: NLG evaluation using gpt-4 with better human alignment," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 2511–2522. [Online]. Available: https://aclanthology.org/2023.emnlp-main.153/

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:51:38 UTC from IEEE Xplore. Restrictions apply.